

Homework 1

Problem 1. Rossby Wave Instability (50 points)

Consider a two-dimensional, inviscid, non-magnetic disk in cylindrical coordinates (r, ϕ) . Neglect disk self-gravity. Let Σ denote surface density and P vertically integrated pressure. Adopt the barotropic equation of state $P = c_s^2 \Sigma$, with constant c_s . The governing equations are

$$\frac{\partial \Sigma}{\partial t} + \nabla \cdot (\Sigma \mathbf{v}) = 0, \quad (1)$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla P}{\Sigma} - \nabla \Phi, \quad \Phi = -\frac{GM_*}{r}. \quad (2)$$

The axisymmetric equilibrium has $v_{r0} = 0$ and $v_{\phi0} = r\Omega(r)$, with

$$\Sigma_0(r) = \Sigma_b \left[1 + A \exp \left(-\frac{(r - r_0)^2}{2w^2} \right) \right], \quad A > 0, \quad w \ll r_0. \quad (3)$$

Define

$$\kappa^2 = \frac{1}{r^3} \frac{d(r^4 \Omega^2)}{dr}, \quad \zeta_0 = \frac{\kappa^2}{2\Omega \Sigma_0}. \quad (4)$$

Consider non-axisymmetric perturbations proportional to $e^{i(m\phi - \omega t)}$, with integer $m \geq 1$. Define the Doppler-shifted frequency $\tilde{\omega}(r) = \omega - m\Omega(r)$ and the enthalpy perturbation $W(r) = \delta P / \Sigma_0$.

1. **Equilibrium and linearization (15 points).** Derive $\Omega^2(r)$ from radial force balance and write the linearized continuity, radial-momentum, and azimuthal-momentum equations. State the conditions needed for $\Omega^2 > 0$ and $\kappa^2 > 0$.
2. **Radial eigenvalue equation (20 points).** Eliminate δv_r , δv_ϕ , and $\delta \Sigma$ to obtain a second-order ordinary differential equation for $W(r)$. Write every coefficient explicitly in terms of background quantities and ω . Show where the vortensity gradient enters the equation and identify the corotation radius r_c , defined by $\text{Re}(\omega) = m\Omega(r_c)$.
3. **Boundary conditions and instability (10 points).** Consider a finite domain $r_{\text{in}} < r < r_{\text{out}}$ enclosing the bump, with impermeable boundaries $\delta v_r = 0$. Translate these conditions into conditions on W . Explain why ω is an eigenvalue and give the growth condition under the stated time convention. Distinguish the role of a vortensity extremum from a sufficient condition for an unstable global mode.
4. **Physical interpretation (5 points).** Explain how Rossby waves on opposite sides of a vortensity extremum can interact. Explain why a density bump alone does not justify assuming that the disk is unstable.

Problem 2. Magnetized VSI in Ideal MHD (50 points)

Use a local Cartesian frame (x, y, z) rotating at angular frequency Ω , with x , y , and z pointing in the radial, azimuthal, and vertical directions. Adopt an inviscid, incompressible local model with uniform reference density ρ_0 and ideal MHD. The density is taken to be spatially uniform and constant in time, with no density perturbations:

$$\boxed{\rho(\mathbf{x}, t) = \rho_0, \quad \delta\rho = 0.}$$

Take the background flow and uniform poloidal magnetic field to be

$$\mathbf{U}_0 = [-q\Omega x + (\partial_z U_{0y})z]\mathbf{e}_y, \quad \mathbf{B}_0 = B_x\mathbf{e}_x + B_z\mathbf{e}_z, \quad B_z \neq 0, \quad (5)$$

where $q = -d \ln \Omega / d \ln R = 3/2$ is the Keplerian radial shear parameter, and $\partial_z U_{0y}$ is the local vertical shear. Treat Ω , q , and $\partial_z U_{0y}$ as constants. The reference point is away from the disk midplane when $\partial_z U_{0y} \neq 0$.

Consider axisymmetric perturbations proportional to $e^{st+i(k_x x + k_z z)}$, with $k_z \neq 0$. Use the definitions

$$\mathbf{u} = \delta\mathbf{v}, \quad \mathbf{b} = \frac{\delta\mathbf{B}}{\sqrt{4\pi\rho_0}}, \quad \Pi = \frac{\delta P}{\rho_0} + \frac{\mathbf{B}_0 \cdot \delta\mathbf{B}}{4\pi\rho_0}, \quad (6)$$

$$\mu = \frac{k_x}{k_z}, \quad \omega_A = \frac{\mathbf{k} \cdot \mathbf{B}_0}{\sqrt{4\pi\rho_0}}, \quad \kappa^2 = 2(2 - q)\Omega^2 = \Omega^2. \quad (7)$$

Here Π is the perturbed total pressure per unit mass.

1. Consistency of the background (10 points). Starting from

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}), \quad \nabla \cdot \mathbf{B} = 0,$$

derive the condition on B_x/B_z for a time-independent background. Explain why a purely vertical field is not a stationary background when $\partial_z U_{0y} \neq 0$. Use a background satisfying the condition you derive for the remaining parts.

2. Linearized equations (15 points). Derive the three velocity and three induction equations, retaining Coriolis forces, radial shear, vertical shear, and magnetic tension. Use $\mathbf{k} \cdot \mathbf{u} = \mathbf{k} \cdot \mathbf{b} = 0$ to eliminate u_z and b_z . Eliminate Π and write the resulting four-variable system for (u_x, u_y, b_x, b_y) .

3. Dispersion relation (15 points). Set the determinant of this system to zero and obtain a polynomial in s . Write its coefficients explicitly in terms of Ω , κ , $\partial_z U_{0y}$, μ , and ω_A . Account for the background-field constraint when interpreting ω_A . Verify the hydrodynamic VSI limit $\omega_A \rightarrow 0$ and the classical vertical-field MRI limit $\partial_z U_{0y} = 0$, $k_x = 0$.

4. Growth and interpretation (10 points). Derive the hydrodynamic VSI growth condition. For the magnetized problem, determine the condition for a real positive root s^2 and discuss whether magnetic tension always stabilizes the flow.

The local magnetized model in Problem 2 is a deliberately simplified ideal-MHD problem, rather than a full vertically global disk model.

Suggested reading.

Lovelace et al. (1999), ApJ, 513, 805. [DOI](#)

Li et al. (2000), ApJ, 533, 1023. [DOI](#)

Balbus & Hawley (1991), ApJ, 376, 214. [DOI](#)

Latter & Papaloizou (2018), MNRAS, 474, 3110. [DOI](#)